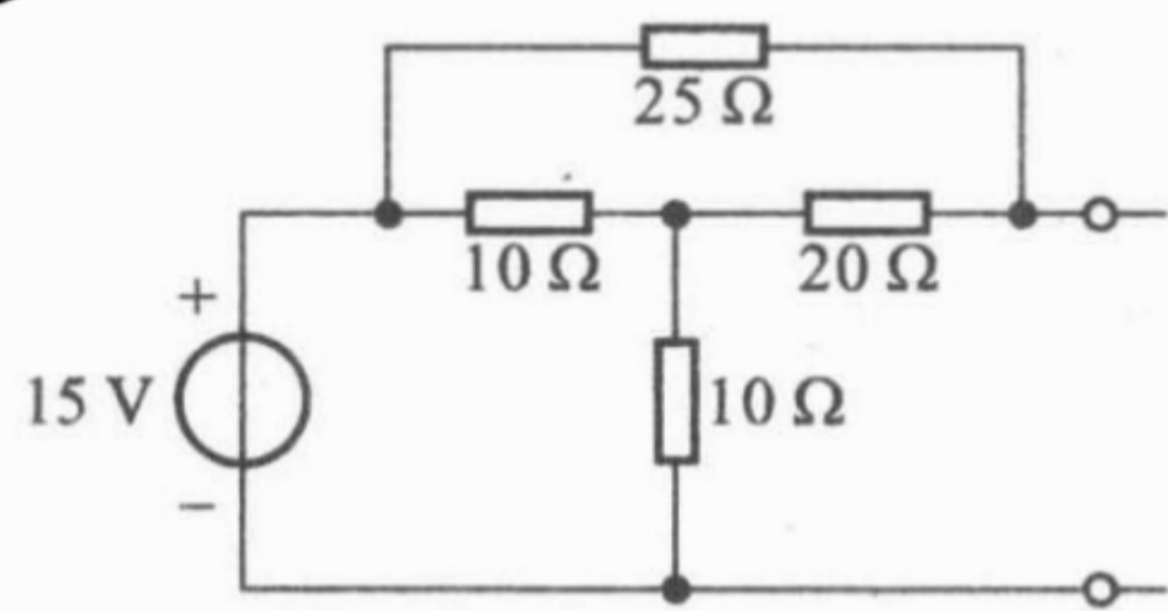
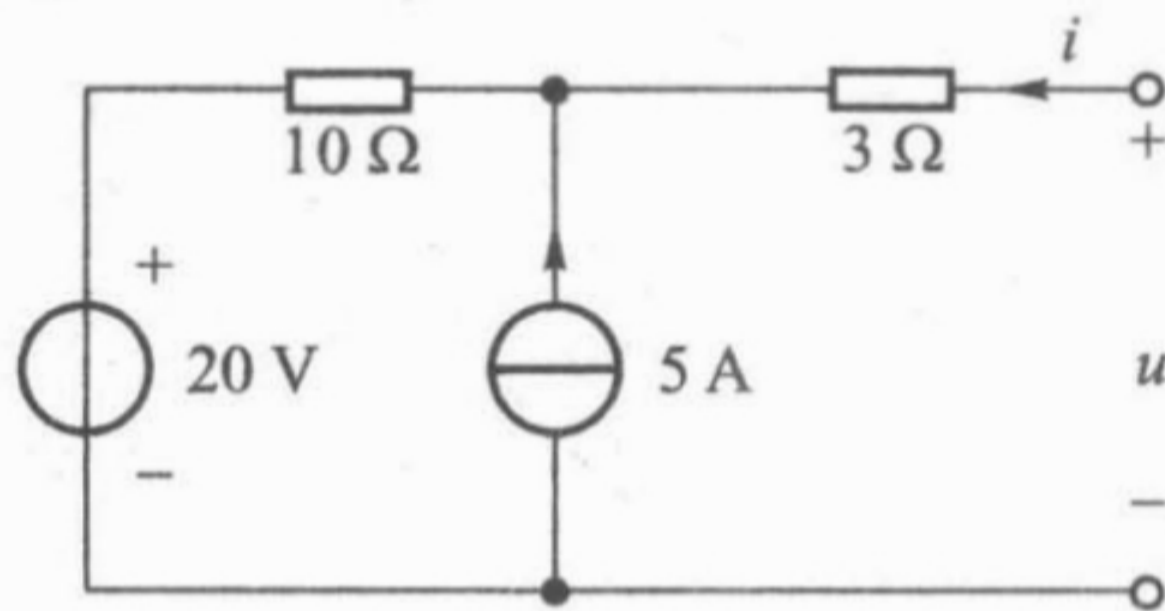


第四章作业.

4-2 试用外施电源法求图题 4-2 所示含源单口网络的 VCR, 并绘出伏安特性曲线。



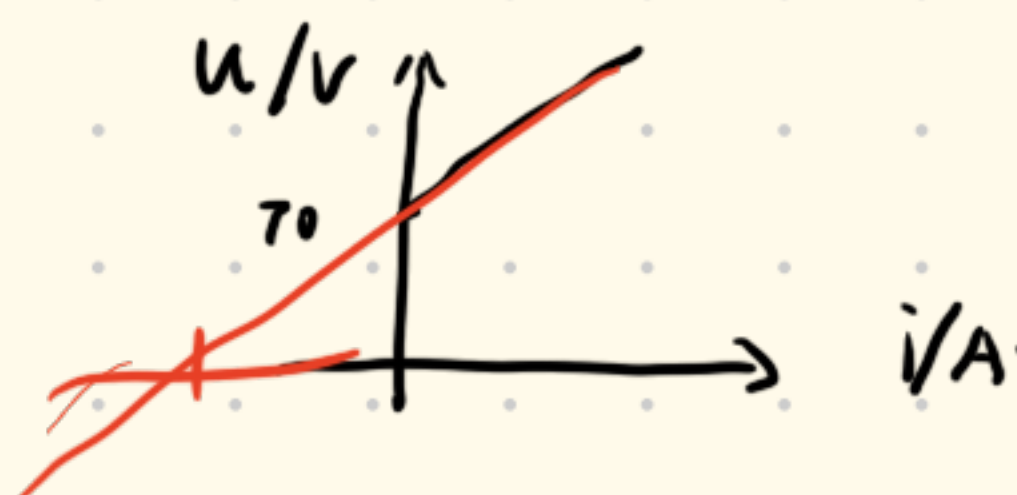
图题 4-1



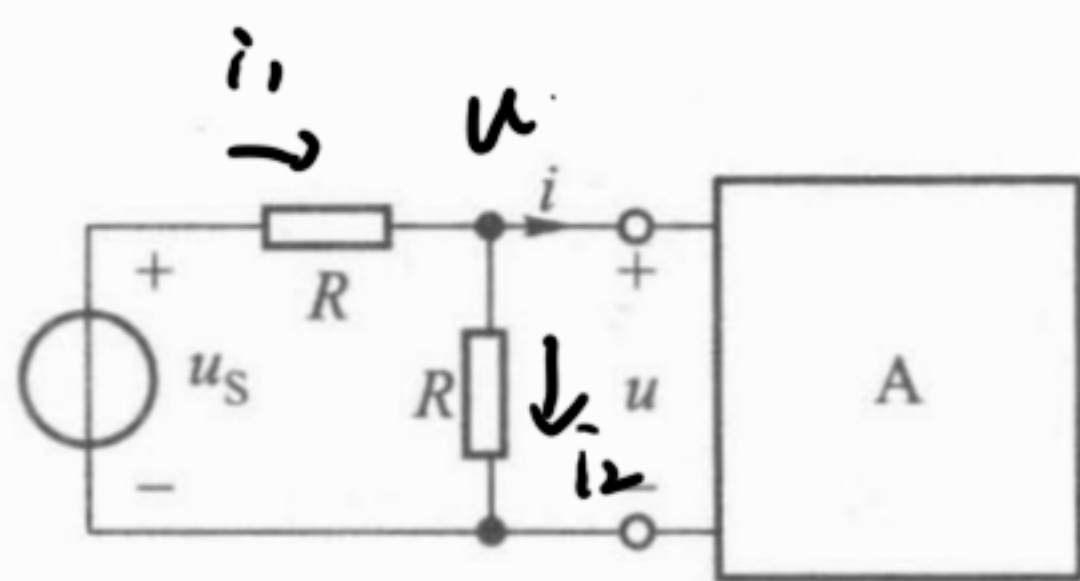
图题 4-2

$$u = 3i + 20 + 10(i + 5) = 13i + 70$$

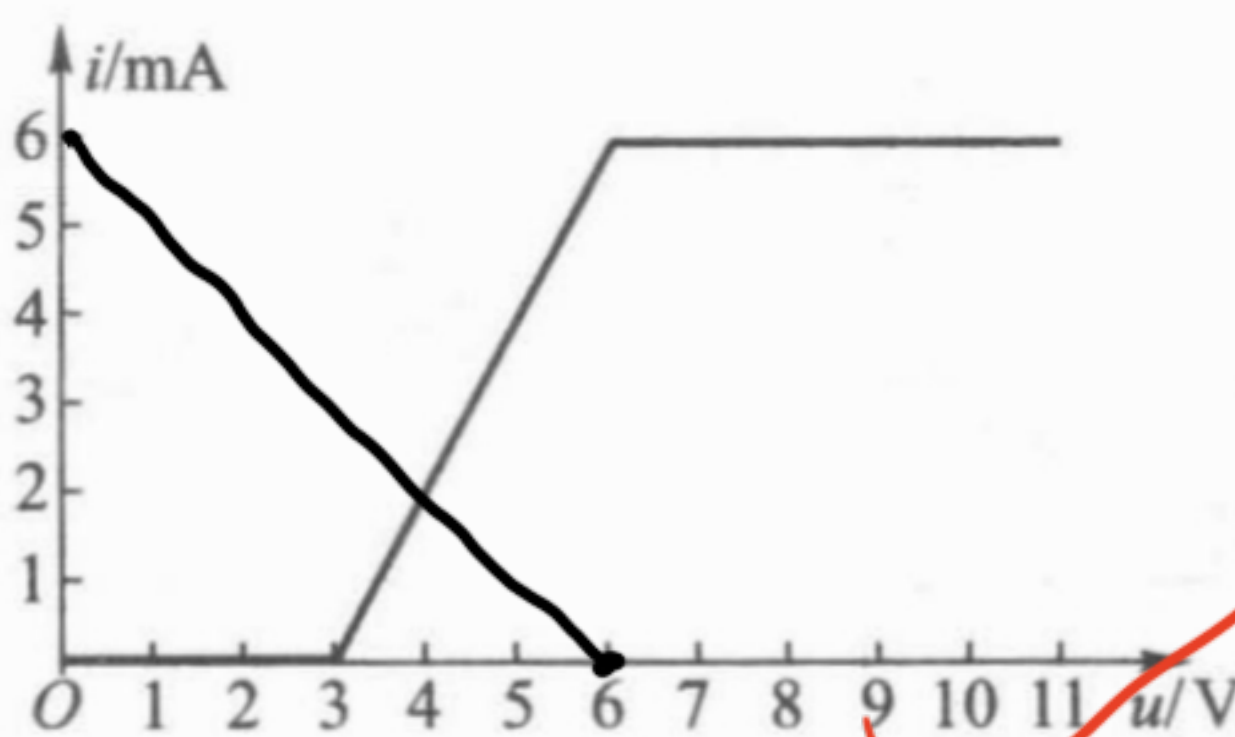
70
-13



4-7 电路如图题 4-7(a) 所示, $u_s = 12\text{ V}$, $R = 2\text{ k}\Omega$ 网络 N 的 VCR 如图(b) 所示, 求 u 和 i , 并求流过两线性电阻的电流。



(a)



(b)

图题 4-7

$$\begin{aligned} \left(\frac{1}{R} + \frac{1}{R}\right)u - \frac{1}{R}u_s &= -i \\ \frac{1}{1000}u - \frac{1}{2000} \times 12 &= -i \\ u - 6 &= -1000i \\ u &= 6 - 1000i \end{aligned}$$

$$\begin{cases} u = 6 - 1000i \\ u = 3 + 500i \end{cases} \Rightarrow \begin{cases} i = 2\text{ mA} \\ u = 4\text{ V} \end{cases}$$

$$\Rightarrow \begin{cases} i_1 = \frac{12-4}{2} = \frac{8}{2000} = 4\text{ mA} \\ i_2 = \frac{4}{2} = 2\text{ mA} \end{cases}$$

190000

§ 4-3 练习题 4-6。

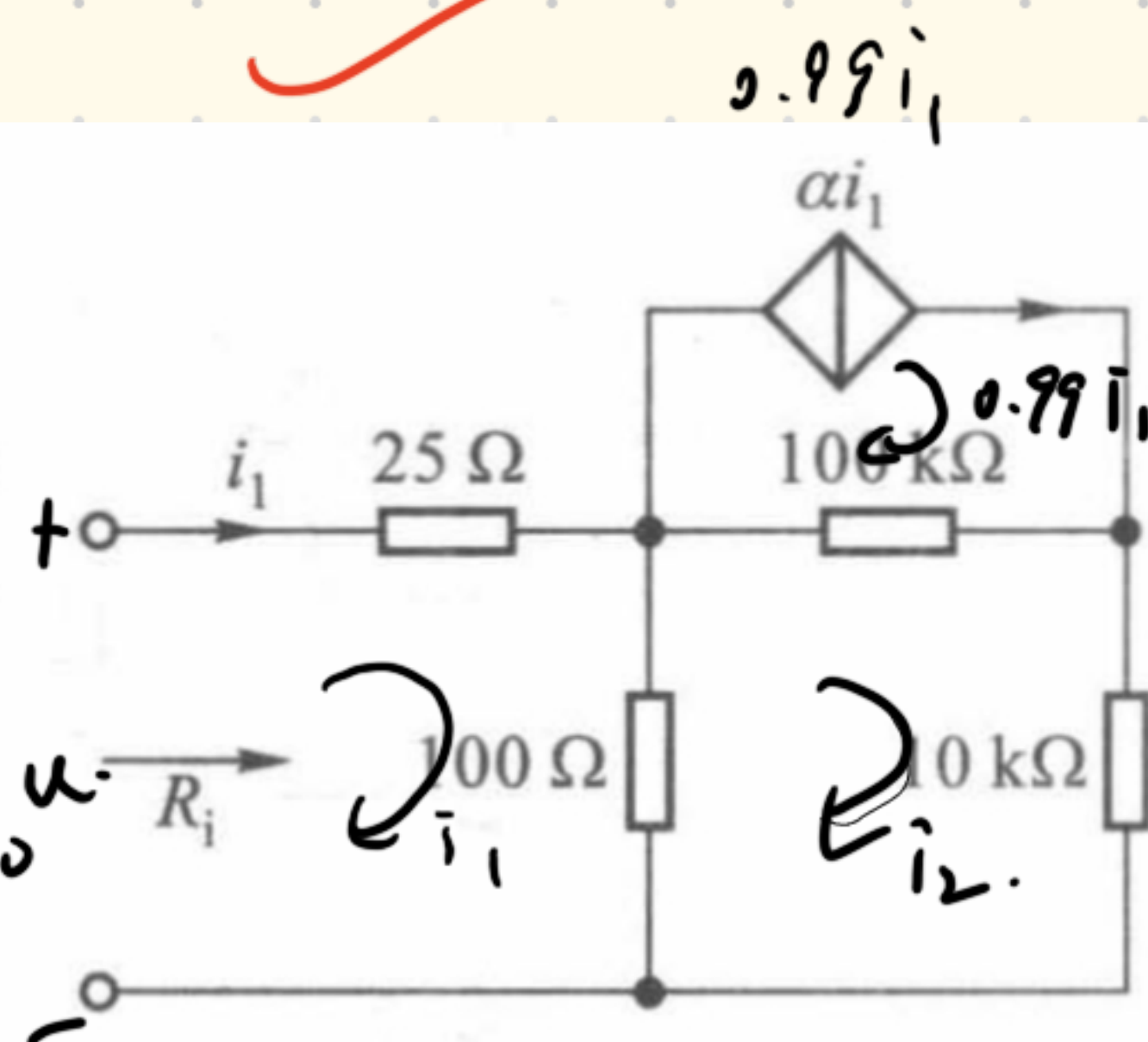
4-10 求图题 4-8 所示电路的输入电阻 R_i , 已知 $\alpha = 0.99$ 。

4-11 对图题 4-9(a) 所含无伴电压源电路, 试证明图(b) 含有伴电压源电路与它是等效的。(称为电压源的

$$u = (25 + 100)i_1 - 100i_2$$

$$10 \times 10^3 i_2 + 100(i_2 - i_1) + 10^5(i_2 - 0.99i_1) = 0$$

$$\Rightarrow \begin{cases} u = 125i_1 - 100i_2 \\ 110100i_2 - 100i_1 - 99 \times 10^3 i_1 = 0 \end{cases}$$

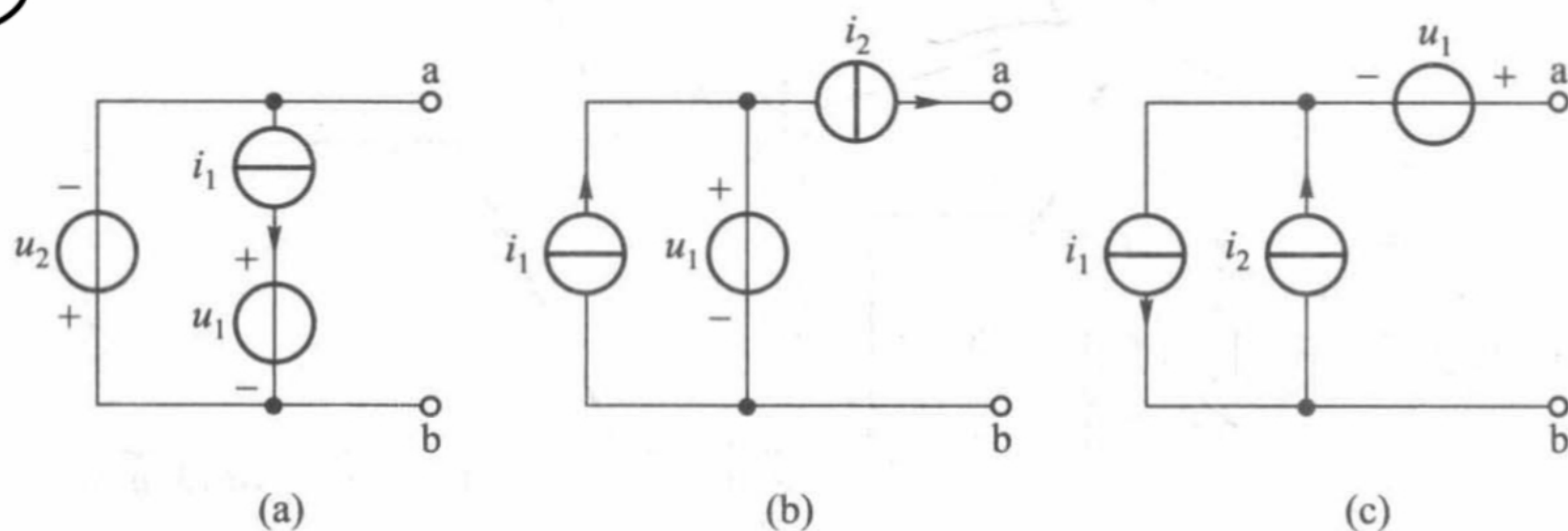


图题 4-8

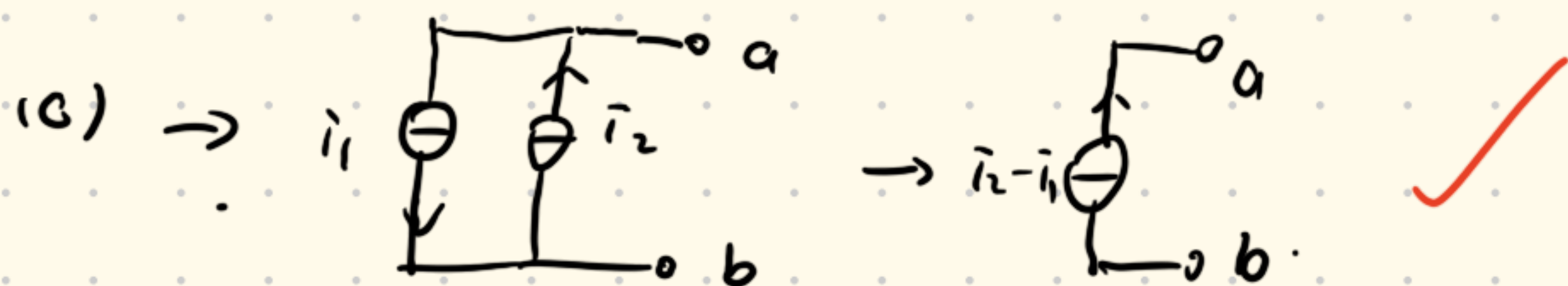
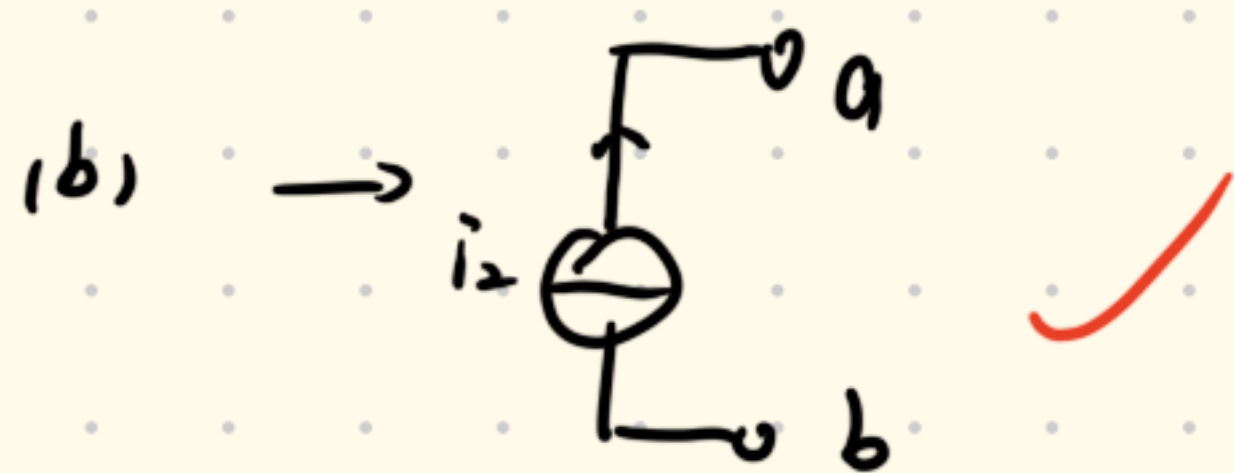
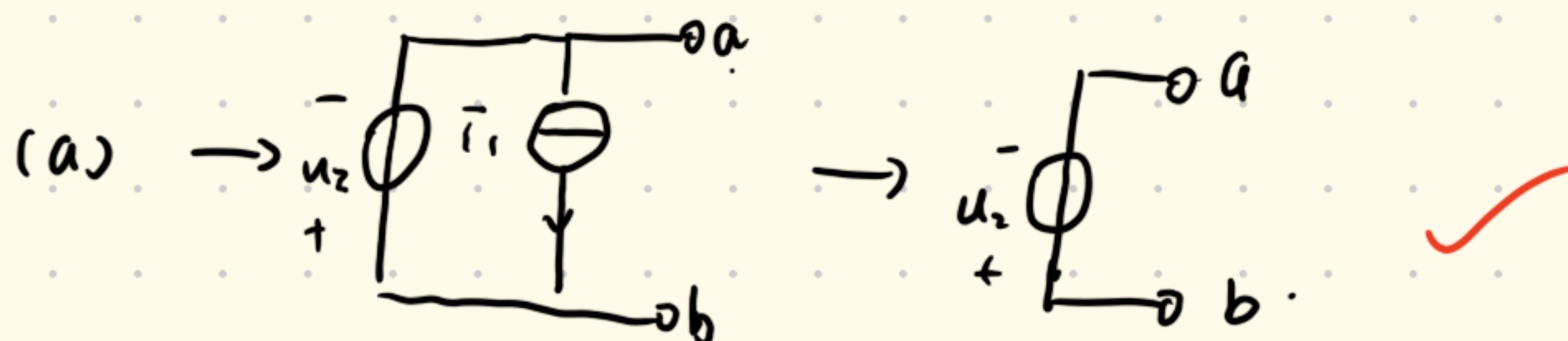
$$\Rightarrow \begin{cases} u = 125i_1 - 100i_2 \\ 110.1i_2 = 99.01i_1 \end{cases} \Rightarrow u = 125i_1 - 100 \times \frac{99.01}{110.1} i_1 = (125 - 100 \times \frac{99.01}{110.1}) i_1$$

$$\therefore Ri = \frac{u}{i_1} = 125 - 100 \times \frac{99.01}{110.1} \approx 35.07 \Omega$$

4-14 试求图题 4-10 所示各电路的等效电路。



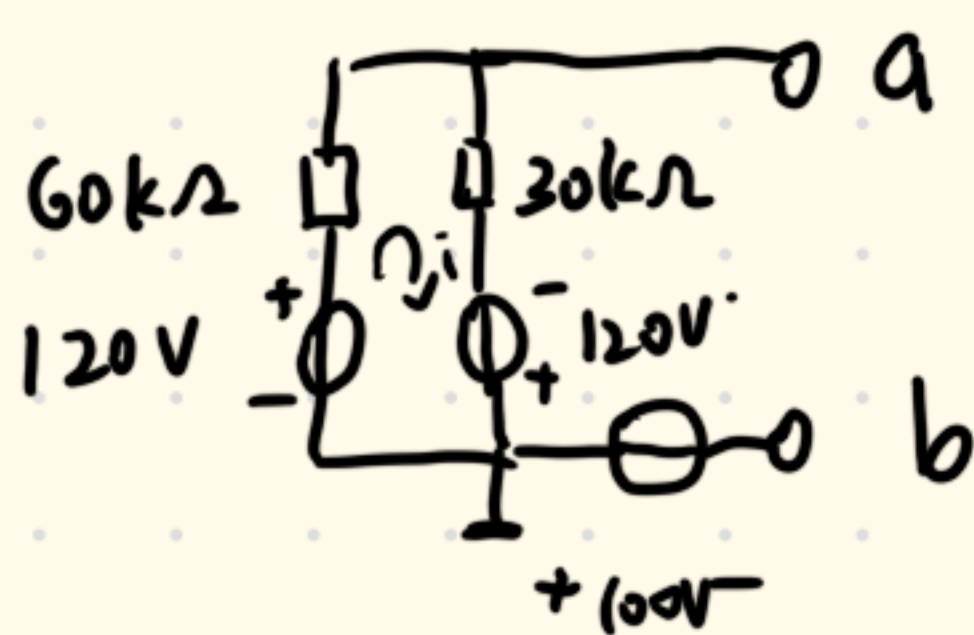
图题 4-10



§ 4-6 戴维南定理

4-16 用戴维南定理求图题 4-11 所示电路中流过 $20 \text{ k}\Omega$ 电阻的电流及 a 点电压 U_a 。

把 $20 \text{ k}\Omega$ 断开



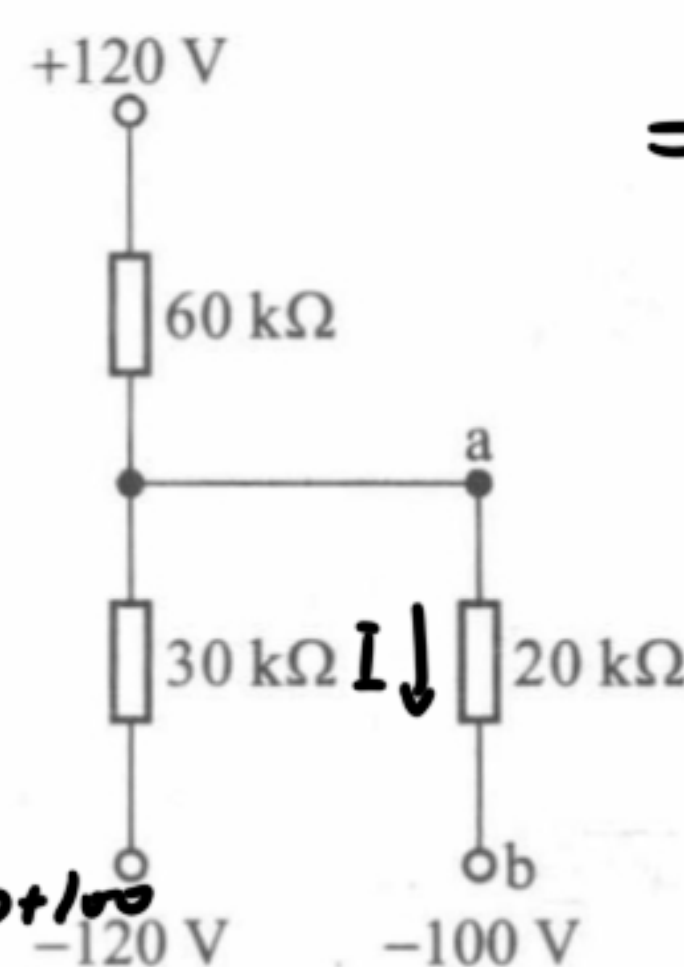
等效电阻

$$R = \frac{60 \times 30}{60 + 30} = 20 \text{ k}\Omega$$

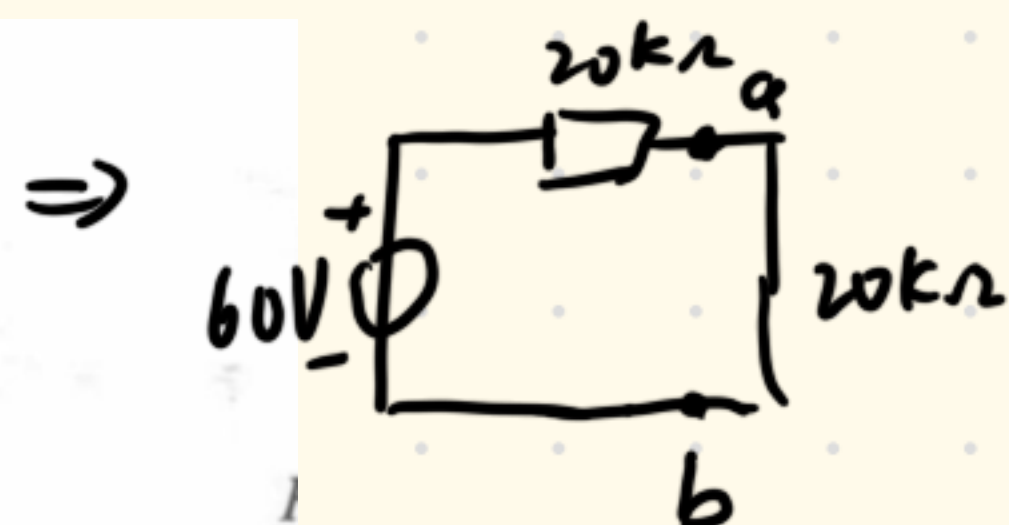
$$(60 \text{ k} + 30 \text{ k}) i = 240$$

$$i = \frac{1}{90 \text{ k}} \times 240$$

$$U_{oc} = \frac{1}{90 \text{ k}} \times 240 \times 30 \text{ k} - 120 + 100 = 60 \text{ V}$$



图题 4-11

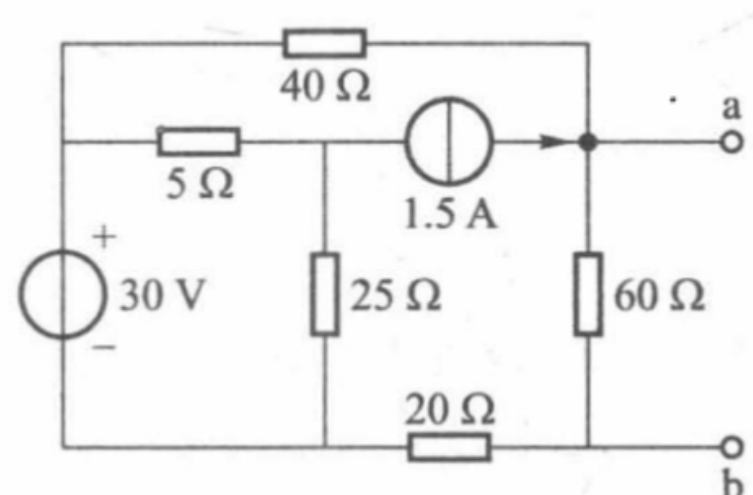


$$I = \frac{60 \text{ V}}{40 \text{ k}\Omega} = 1.5 \text{ mA}$$

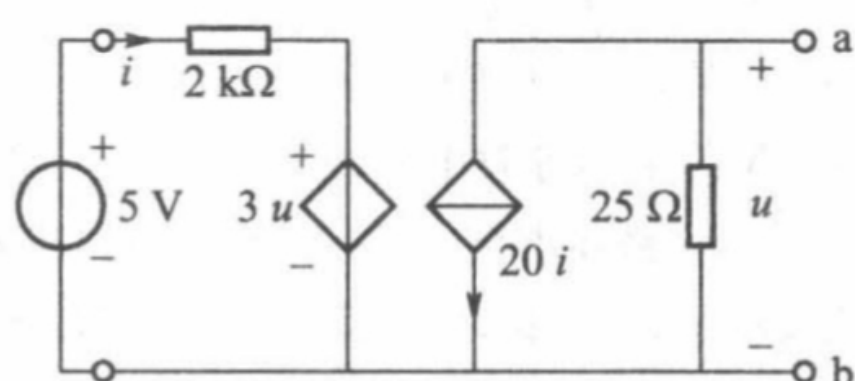
$$U_a = -100 + 60 - 20 \times 1.5 = -70 \text{ V}$$

4-27 (1) 求图题 4-19 电路 ab 端的戴维南等效电路, 求等效电阻指定用 (4-29) 式。
(2) 求等效电阻指定用外施电源法。

$$R_o = \frac{u_{oc}}{i_{sc}} \quad (4-29)$$



图题 4-18



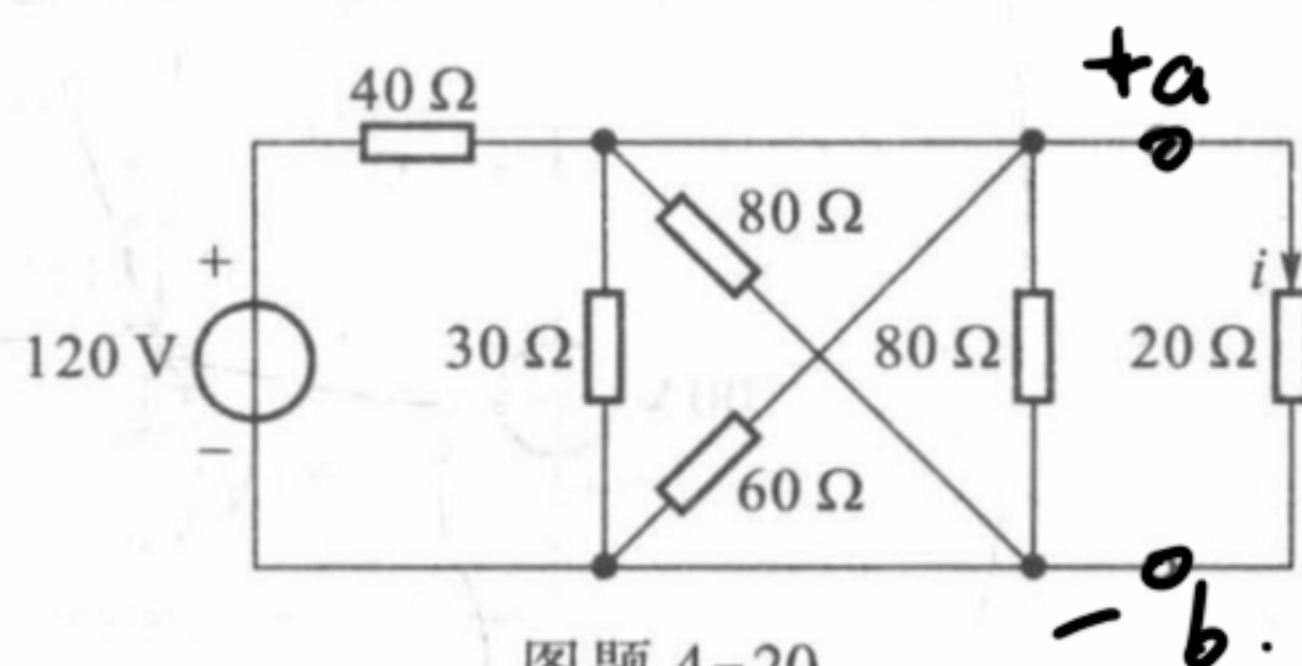
图题 4-19

$$\begin{cases} u_{oc} = (-20i) \cdot 25 = -500i = u \\ i = \frac{5-3u}{2k\Omega} = \frac{5+1500i}{2000} \Rightarrow u = -5V = u_{oc} \end{cases}$$

a、b 短路 $i_{sc} = -20i$
 $u = 0$
 $i = \frac{5V}{2k\Omega} = 2.5mA$
 $i_{sc} = -0.05A$

$$R = \frac{u_{oc}}{i_{sc}} = \frac{-5}{-0.05} = 100\Omega$$

4-28 用诺顿定理求图题 4-20 所示电路的电流 i 。



图题 4-20



$$i_{sc} = \frac{120V}{40\Omega} = 3A$$

去掉电压源:



$$\frac{1}{R} = \frac{1}{40} + \frac{1}{30} + \frac{1}{60} + \frac{1}{80} + \frac{1}{80} = \frac{1}{20} + \frac{1}{30} + \frac{1}{60} = \frac{14+3}{60} = \frac{1}{10}$$

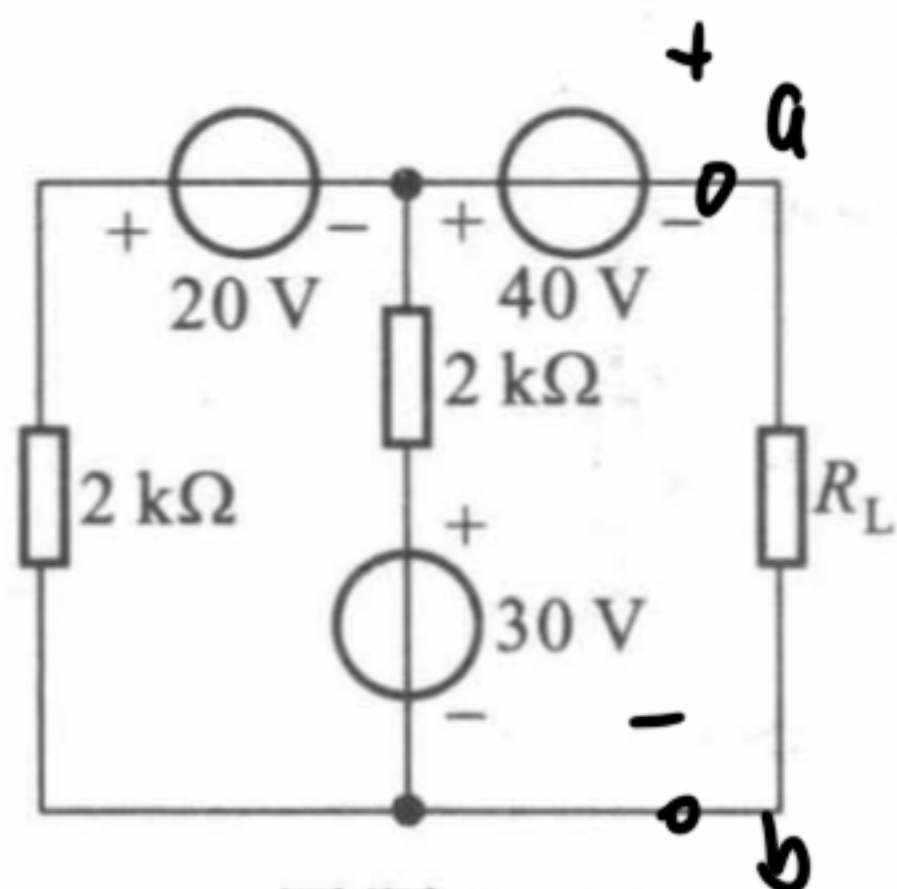
$$R = 10\Omega$$



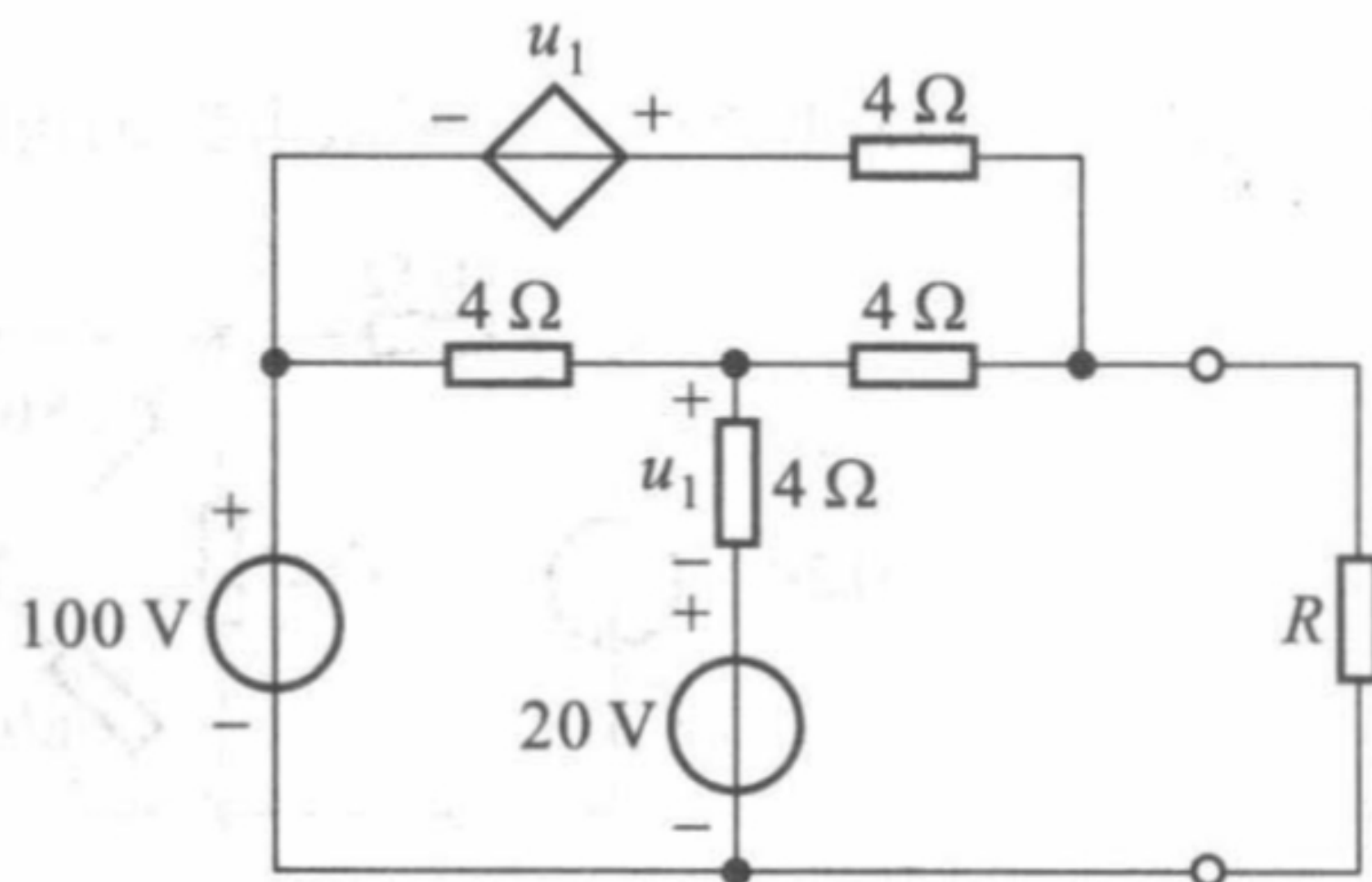
$$i = \frac{10}{30} \times i_{sc} = 1A$$

4-32 电路如图题 4-24 所示。(1) 若 $R_L = 3 \text{ k}\Omega$, 试求 R_L 获得的功率;(2) 求 R_L 能获得的最大功率;(3) 试求 R_L 获得 20 mW 时的电阻值。

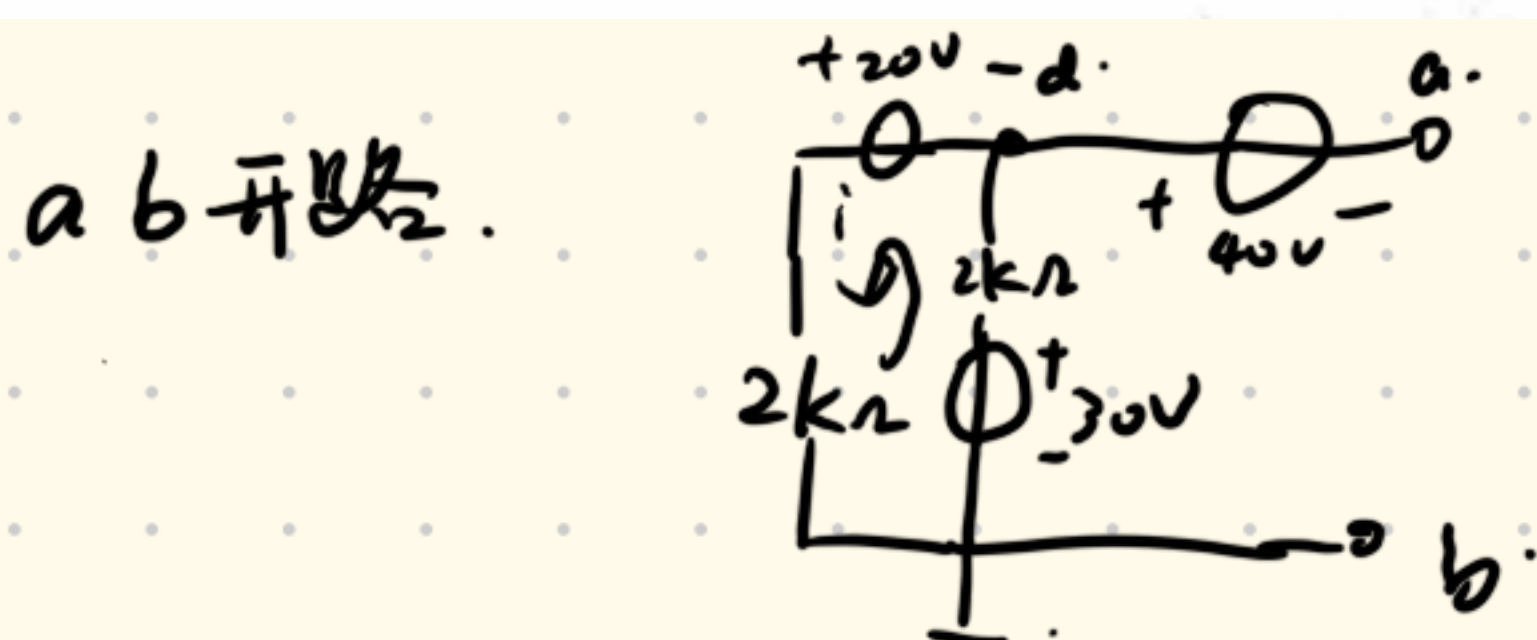
4-33 电路如图题 4-25 所示。(1) 求 R 获得最大功率时的数值;(2) 求在此情况下, R 获得的功率;(3) 求 100 V 电源对电路提供的功率;(4) 求受控源的功率;(5) R 所得功率占电路内电源产生功率的百分比。



图题 4-24



图题 4-25

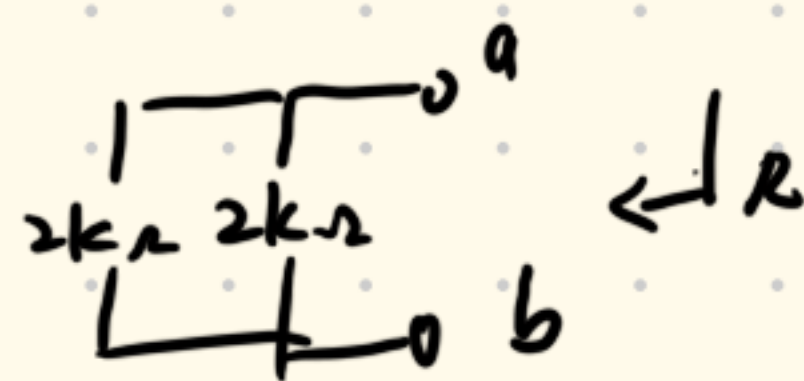


$$2000i - 20 + 2000i - 30 = 0$$

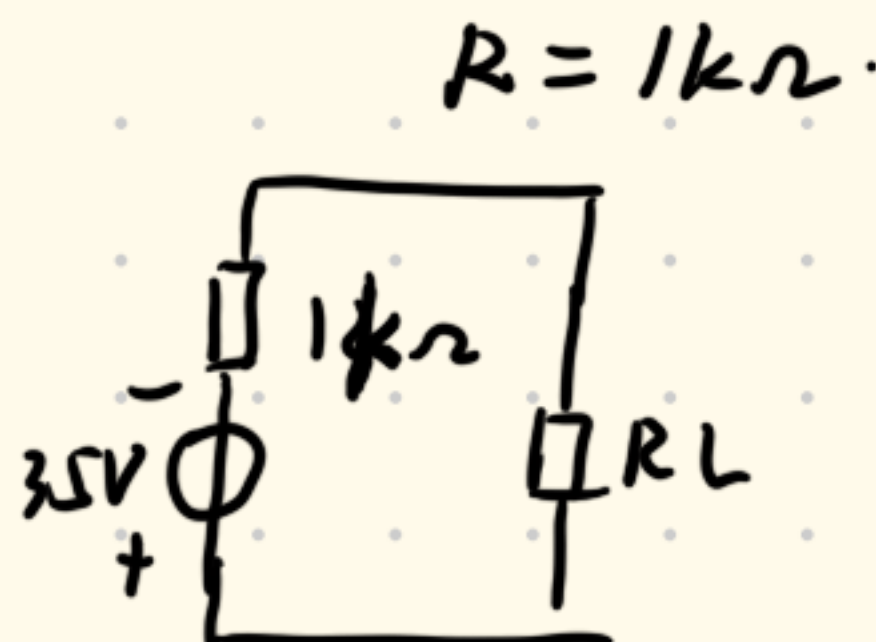
$$i = 12.5 \text{ mA}$$

$$U_{OC} = U_{ab} = 30 - 25 - 40 = -35 \text{ V}$$

戴维南电压源



戴维南等效:



(1) 若 $R_L = 3 \text{ k}\Omega$.

$$I = \frac{35 \text{ V}}{4 \text{ k}\Omega} = \frac{35}{4} \text{ mA}$$

$$P_L = I^2 \cdot R_L = \frac{35}{4} \times \frac{35}{4} \times 10^{-6} \times 3 \times 10^3 = 229.6875 \text{ mW}$$

(2) 若 $R_L = R_{eq}$.

P_L 取得 max 值

$$P_{L_{max}} = \frac{(\frac{35}{2})^2}{1 \text{ k}\Omega} = 306.25 \text{ mW}$$

$$(3) P_L = \left(\frac{U}{1 \text{ k}\Omega + R_L} \right)^2 \cdot R_L = 20 \text{ mW}$$

设 $R_L = X \text{ k}\Omega$.

$$\left(\frac{35}{1+X} \times 10^{-3} \right)^2 \cdot X \cdot 10^3 = 20 \times 10^{-3}$$

$$\Rightarrow \frac{35^2}{(1+X)^2} \cdot X = 20$$

$$\Rightarrow \frac{35^2}{20} = \frac{1+2X+X^2}{X}$$

$$\Rightarrow X^2 - 59.25X + 1 = 0$$

$$X = \frac{59.25 \pm \sqrt{50}}{2}$$

$$\therefore R_L = 16.88 \Omega$$

$$R_L = 59.25 \text{ k}\Omega$$